

Kubernetes AI Cluster Architecture

Workspace: New ks8 project | Based on the last verified cluster state | 15 September 2026

Status legend: RUNNING / AVAILABLE CONFIGURED / CONTROLLER-DEPENDENT BLOCKED / INCOMPATIBLE

Client and access layer			
kubectl / PowerShell Cluster administration, rollout checks, port-forwarding, and model commands. CONFIGURED	Gateway API GatewayClass, Gateway, and HTTPRoute resources are installed. CONFIGURED	Envoy Gateway Referenced controller is not installed, so Gateway status remains Unknown. BLOCKED	HTTP /v1 route Route points to the InferencePool on path prefix /v1. CONFIGURED

Client request → Gateway / HTTPRoute → InferencePool

Model serving layer			
Ollama Deployment ollama-worker, 2 replicas, host networking, port 11434. RUNNING	Llama 3.2 Model downloaded into one Ollama pod; storage is not persistent/shared. INSTALLED	InferencePool ollama-inference-pool selects app=ollama-engine. CONFIGURED	Inference runtime Ollama inference hit illegal instruction because nodes expose no AVX/AVX2. BLOCKED

Ilm-d control plane			
ModelService controller Helm release ilm-d, namespace ilm-d. RUNNING	Redis Installed as ilm-d support infrastructure. RUNNING	API compatibility Chart 1.0.23 expects legacy inference.networking.x-k8s.io/v1alpha2. CRASHLOOP	vLLM workload Requires GPU-capable nodes and a compatible inference API/controller stack. NOT RUNNING

Kind / Docker Desktop cluster

Control plane Kind control-plane node.	CLUSTER	Worker 1 CPU node; no <code>nvidia.com/gpu</code> .	READY	Worker 2 CPU node; no <code>nvidia.com/gpu</code> .	READY	Hardware boundary Docker Desktop nodes expose no AVX/AVX2 and no NVIDIA GPU.	LIMITATION
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Component inventory and operational paths

These are the main objects and the path each request would take when all controllers are available.

Ollama path • Deployment: ollama-worker • Service: ollama-service:11434 • Model: llama3.2:latest • Enterprise TLS CA mounted through a ConfigMap. • Direct Ollama API after CPU compatibility is resolved.	llm-d path • Helm release: llm-d-1.0.23 • ModelService CRD installed. • ModelService: llama3-vllm • Redis support pod is running. • Standard path targets vLLM, not Ollama.
Routing path • GatewayClass: ollama-gateway-class • Gateway: llm-d-gateway • HTTPRoute: llmd-ollama-route • Gateway controller is absent, so resources are not programmed.	Required changes for a fully active platform • Start Docker Desktop / Kubernetes. • Install Envoy Gateway or kgateway. • Align inference-extension CRD versions with llm-d chart version. • Use GPU nodes for standard vLLM ModelService. • Use persistent shared storage for model artifacts.

Important status note: The Kubernetes API was unreachable when this document was generated, so the diagram reflects the last verified state from the cluster. It separates configured resources from components that were actually running.

Suggested validation commands

docker info and **kubectl get nodes** confirm cluster availability.
kubectl get pods -A -o wide shows running workloads.
kubectl get gatewayclass,gateway,httproute -A shows Gateway programming state.
kubectl get modelservice,inferencepool -A shows llm-d resources.
kubectl get pods -n llm-d shows controller and Redis health.